

Hanwool Jang

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Employment

Glasgow Caledonian University
Department of Finance, Accounting and Risk
Lecturer (Assistant Professor), 2022 - Present
Program Leader, MSc Financial Technology, 2025 - Present

Yonsei University
Department of Industrial Engineering
Postdoctoral Fellow, 2020 - 2021
Department of Investment Information Engineering
Instructor, 2020 - 2021
Center for Finance and Technology
Research Director, 2021

Education

KAIST Moon Soul Graduate School of Future Strategy
PhD in Engineering, 2018 - 2020
MA in Management Studies, 2016 - 2018

Tsinghua University, School of Mechanical Engineering
BE in Vehicle Engineering, 2006 - 2010

Research Interests

Quantitative Finance, Real Estate, Applications of AI and Quantum Computing

Publications

Asymmetric impacts of artificial intelligence on housing price valuation across education levels
with An S, Song Y, Ahn K, Humanities and Social Sciences Communications, 2025

Toward transparent and accurate housing price appraisal: Hedonic price models versus machine learning algorithms
with An S, Song Y, Ahn K, Financial Innovation 1, 141, 2025

Impact of monetary policy on corporate defaults and associated welfare costs
with Ahn K, Lee D, Computational Economics, 2025

The impact of futures trade on the informational efficiency of US REIT market
with Jeong M, Ahn K, Sohn S, Financial Innovation 11(45), 2025

CEO turnover and financial policy transfer
with Kim D, Ahn K, Lee J, North American Journal of Economics and Finance, 77, 2025

Business cycle and herding behavior in stock returns: Theory and evidence
with Cong L, Kim D, Ahn K, Financial Innovation 10(6), 2024
Featured in Phys.org: Quantum mechanics model unveils hidden patterns in stock markets

The effect of rare events on information-leading role: Evidence from real estate investment trusts and overall stock markets
with An S, Choi G, Ahn K, Humanities and Social Sciences Communications 11, 1628, 2024

Assessment of street-level greenness and its associations with housing prices in metropolitan area

- with An S, Song Y, Kim H, Ahn K, Scientific Reports 13, 22577, 2023
- COVID-19 and REITs crash: Predictability and market conditions
with Ryu I, Ahn K, Kim J, Computational Economics, 2023
- Deep neural network as a tool for appraising housing prices: A case study of Busan, South Korea
with An S, Song Y, Ahn K, Journal of Physics: Conference Series 2287, 2022
- Fixed rate mortgages: The cost of interest rate risk aversion
with Forsyth J, Kim D, Ahn K, Finance Research Letters 44, 102158, 2022
- Market efficiency of US REITs: Revisit
with Ryu I, Kim D, Ahn K, Chaos, Solitons & Fractals 150, 111070, 2021
Featured in University News: Academic Insight Dr Hanwool Jang on financial market efficiency
- Can government stabilize the housing market? Evidence from South Korea
with Song Y, Ahn K, Physica A: Statistical Mechanics and its Applications 550, 124114, 2020
- Economic impacts of being close to subway networks: A case study of Korean metropolitan areas
with Song Y, Ahn K, Research in Transportation Economics 83, 100900, 2020
- Are referees unfair to away teams? Evidence from the Premier League
with Choi G, Kim D, Ahn K, Springer Proceedings in Mathematics and Statistics 446, 105–111, 2024
- Hedonic dataset of the metropolitan housing market – cases in South Korea
with An S, Ahn K, Song Y, Data in Brief, 35, 106877, 2021
- Associations of transport accessibility with restaurants' survival in Gwangju Metropolitan Area
with Song Y, Lee K, Journal of the Economic Geographical Society of Korea 23(4), 425-437, 2020
(In Korean)
- Accessibility and spatial equity of subway networks in Seoul
with Song Y, Lee K, Journal of the Economic Geographical Society of Korea 22(4), 513-525, 2019
(In Korean)

Research Grants

Principal Investigator

- Quantum Technologies Alliance for Research Challenge, UK (£10,000)
Scotland-Korea collaboration on quantum simulation in asset pricing based on the quantum harmonic oscillator, 2025
- Postdoctoral Fellowship for Next-Generation Researchers in Science and Engineering, Korea (\$ 81,000)
Real estate pricing system: AI and Econometrics driven platform, 2020 - 2022
- Global PhD Fellowship, Korea (\$ 54,000)
FRM vs. ARM: The optimal choice, 2019 - 2020

Investigator

- Future Strategy Research, Korea (\$ 50,000)
Bubble observatory, Korea, 2018
- 4th Industrial Revolution and Convergence Research (\$ 26,400)
Predicting the critical time of financial bubble, 2017

Teaching Experience

- Corporate Finance & Financial Strategies
- Fundamentals of Corporate Finance
- Strategic Bank Management
- Bank Management in Global Context

Financial Risk Modelling
 Introduction To Quantitative Data Analysis
 AI and Machine Learning
 Data Visualisation for Business Analytics
 Seminar in Investment Information Engineering 2
 Principles of Financial Engineering

Seminar/ Conference Presentations

2026

University of Westminster Visit Programme, Glasgow Caledonian University, UK
 Kongju National University (Online), Korea

2025

Konkuk University (Online), Korea

2024

Quantum x Finance workshop, University of Glasgow, UK
 Risk Society & Governance Research Seminar (Online), Glasgow Caledonian University, UK

2023

AsRES-GCREC & AREUEA International Real Estate Joint Conference, Hsinchu, Taiwan
 Risk Society & Governance Research Seminar (Online), Glasgow Caledonian University, UK
 12th International Conference on Mathematical Modeling in Physical Sciences, Serbia
 Chung-Ang University (Online), Korea
 Chonnam National University, Korea

2022

Korean Operations Research and Management Science Society, Korea
 Korean Physical Society: Spring Meeting (Online), Korea
 Korean Institute of Industrial Engineering: Spring Meeting, Jeju, Korea

2021

Glasgow Caledonian University (Online), UK
 University of Manchester (Online), UK
 University of Aberdeen (Online), UK
 National Taiwan Normal University (Online), Taiwan
 Korean Institute of Industrial Engineering: Fall Meeting (Online), Dongguk University, Korea
 Institute of Convergence Science Forum (Online), Yonsei University, Korea
 Mathematical Finance Theory, (Online), Yonsei University, Korea
 Reliability Engineering, (Online), Yonsei University, Korea
 Technology Exchange, Wonkwang University, Korea

2020

Korean Association of Regional Geographers: Fall Meeting (Online), Korea
 Korean Physical Society: Spring Meeting (Online), Korea

2019

AREUEA International conference, Bocconi University, Italy
 Thredbo, Nanyang Technological University, Singapore
 Asian Real Estate Society Annual Meeting, Shenzhen, China
 Korean Association of Regional Geographers, Catholic University of Daegu, Daegu, Korea
 Department of Investment Information Engineering, Yonsei University, Seoul, Korea
 Reliability Engineering, Yonsei University, Seoul, Korea

2018

Association of American Geographers Annual Meeting, New Orleans, USA

International Conference on Computational Science, Wuxi, China
Asian Real Estate Society Annual Meeting, Incheon, Korea
The Economic Geographical Society of Korea, Gwangju, Korea
Annual Meeting of Korean Geographers, Seoul, Korea
Graduate School of Future Strategy, KAIST, Daejeon, Korea
Korea Research Institute for Human Settlements, Sejong, Korea
Graduate School of Future Strategy, KAIST, Daejeon, Korea

2017

Physics in Economic and Social System, KAIST, Daejeon, Korea

Honors and Awards

GCU Teaching Awards Nominee, 2024
AFA Annual Meeting Travel Grant, 2020
KAIST Scholarship, 2016 - 2020

Services

Reviewer

International Review of Financial Analysis, Computational Economics, Finance Research Letters, Long Range Planning, Financial Innovation, Chaos Solitons & Fractals, Research in Transportation Economics, Applied Economics Letters, Economies, Transportation, PLoS ONE

Other Employment

Sales Assistant Manager, Division of Power Distribution System, LS Electric, 2016
Mechanical Design Engineer, Division of Electric Vehicle Power Control System, LS Electric, 2013 - 2015
Serve in the Military, Korea Army, 2010 - 2012

Skills

Languages

Korean, Fluent in English and Chinese

Computer Languages and Statistical Packages

C, MATLAB, STATA, Python, R

References

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